

Primary Objective

By the end of this lesson, I will be able to...

- classify a real number into *every* set it belongs to, and explain why a radical sign does not automatically make a number irrational;
- name the *properties of real numbers* (commutative, associative, distributive, identity, inverse, zero) and the *properties of equality* and *inequality*, and say which kind changes one side and which changes both;
- *justify* every line of a solved equation or inequality by naming the property that allowed it — including the rule that reverses an inequality symbol.

Vocabulary & Key Concepts

Fill in each term as we build it together.

Rational number:

A number that can be written as $\frac{a}{b}$ where a and b are integers and $b \neq 0$; as a decimal it terminates or repeats.

Irrational number:

A real number that cannot be written as such a ratio; its decimal never terminates and never repeats.

Examples: π , $\sqrt{5}$.

Closure:

A set is closed under an operation when performing that operation on members of the set always gives an answer still inside the set.

Property of real numbers:

A rule for rewriting one expression as an equal expression — it changes how something looks, not what it is. Only one side is involved.

Property of equality:

A rule that produces a new equation with the same solutions by doing the same thing to both sides at once.

Property of inequality:

The same idea for $<$, $\leq$, $>$, $\geq$ — with one exception: multiplying or dividing both sides by a negative number reverses the symbol.

Counterexample:

A single case that makes a general claim false. One is enough to disprove a rule.

Hook

Two students solved the *same* inequality, $-3x > 12$, and got different answers.

Ana's answer: $x > -4$

Ben's answer: $x < -4$

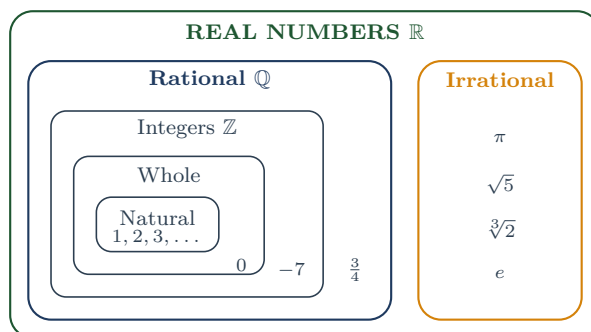
Exactly one of them is right. Don't answer from memory — figure out how we could **settle** it.

- Test $x = -10$ in the *original* inequality. Is $-3(-10) > 12$ true? **yes, $30 > 12$**
Whose answer contains -10 ? **Ben's**
- Test $x = 0$. Is $-3(0) > 12$ true? **no, $0 > 12$ is false** Whose answer contains 0 ? **Ana's**
- So the correct answer is **Ben's, $x < -4$** , and the other student forgot to **reverse the inequality symbol when dividing both sides by -3 , a negative number.**.....

Testing settled *this* one. But we cannot test every problem for the rest of the year. What we need is the **rule** — and its name. *Every legal move in algebra has a name. Today we hand out all the names.*

1. The Real Number System — and Why It Kept Growing

Each set of numbers was invented because the one before it **could not answer a question**.



- **Natural** numbers count things: $1, 2, 3, \dots$. They cannot express “none,” so we add 0 to get the **whole** numbers.
- Whole numbers cannot answer $3 - 5$, so we add the negatives to get the **integers**.
- Integers cannot answer $3 \div 5$, so we allow ratios and get the **rationals**: any number that can be written $\frac{a}{b}$ with a, b integers and $b \neq 0$. As a decimal, a rational number always **terminates** or **repeats**.
- Rationals cannot answer “what is the side length of a square with area 5 ,” so the **irrationals** fill in the gaps. Their decimals never terminate and never repeat.
- Together, the rationals and irrationals make the **real numbers** — with no number in both and none left over.

The big idea behind the whole picture. A set is **closed** under an operation if performing that operation on members of the set always gives you an answer *still inside* the set. Each new set above was invented because the one before it was *not closed* under some operation.

Two traps. A number belongs to *every* set that contains it — so -7 is an integer **and** a rational **and** a real. And the way a number is *written* is not what decides:

$$\sqrt{9} = 3, \text{ so it is really a natural no. } \frac{8}{2} = 4, \text{ so it is really an integer } \sqrt{5} \text{ is irrational}$$

So the radical sign is **not** the test. The **value** is.

2. Properties of Real Numbers — Rewriting *One* Expression

These rules let you change how an expression **looks** without changing what it **is**. Nothing here is “done to both sides” — there are no sides yet.

Property	In symbols	Example
Commutative (+)	$a + b = b + a$	$3 + 8 = 8 + 3$
Commutative ($\times$)	$ab = ba$	$5 \cdot 4 = 4 \cdot 5$
Associative (+)	$(a + b) + c = a + (b + c)$	$(2 + 6) + 9 = 2 + (6 + 9)$
Associative ($\times$)	$(ab)c = a(bc)$	$(3 \cdot 5) \cdot 2 = 3 \cdot (5 \cdot 2)$
Distributive	$a(b + c) = ab + ac$	$4(x + 7) = 4x + 28$
Additive identity	$a + 0 = a$	$-6 + 0 = -6$
Multiplicative identity	$a \cdot 1 = a$	$\frac{2}{5} \cdot 1 = \frac{2}{5}$
Additive inverse	$a + (-a) = 0$	$11 + (-11) = 0$
Multiplicative inverse	$a \cdot \frac{1}{a} = 1, a \neq 0$	$\frac{3}{4} \cdot \frac{4}{3} = 1$
Multiplicative property of zero	$a \cdot 0 = 0$	$(x + 9) \cdot 0 = 0$
Zero product property	if $ab = 0$ then $a = 0$ or $b = 0$	if $(x - 2)(x + 5) = 0$ then $x = 2$ or $x = -5$

Careful — **these are only for + and $\times$** . Commutative and associative say nothing about subtraction or division. Prove it yourself with one example each:

Is $10 - 4$ the same as $4 - 10$? $10 - 4 = 6$ but $4 - 10 = -6$, so subtraction is **not commutative** Is $12 \div 3$ the same as $3 \div 12$?

A single example that breaks a rule is called a **counterexample** — and one is all you ever need.

Two notes worth keeping. The **multiplicative inverse** rule says $a \neq 0$, because zero has no reciprocal — and that is exactly why dividing by **0** is undefined. And the **zero product property** is the reason factoring will solve equations for you all year: zero is the only number with that behavior, which is why we always move everything to one side and set it equal to **0** first.

3. Properties of Equality — Changing *Both* Sides

The distinction that runs this whole course:

Properties of *real numbers* rewrite **one** expression.

Properties of *equality* act on **both sides at once** to make a new equation.

An equation is a claim that the two sides are **two names for the same number**. Everything below follows from that one idea.

Property	In symbols	What it lets you do
Reflexive	$a = a$	Any number equals itself.
Symmetric	if $a = b$ then $b = a$	Write $x = 5$ or $5 = x$ — your choice.
Transitive	if $a = b$ and $b = c$ then $a = c$	Chain two equalities together.
Substitution	if $a = b$, a may replace b anywhere	Swap a name for another name of the same number.
Addition	if $a = b$ then $a + c = b + c$	Add the same amount to both sides.
Subtraction	if $a = b$ then $a - c = b - c$	Subtract the same amount from both sides.
Multiplication	if $a = b$ then $ac = bc$	Multiply both sides by the same amount.
Division	if $a = b$ then $\frac{a}{c} = \frac{b}{c}$, $c \neq 0$	Divide both sides by the same <i>nonzero</i> amount.

Now justify a solution, line by line. Name the property beside each step. Watch how the two *kinds* of property alternate.

Step	Property that allowed it
$3(x - 4) = 18$	Given
$3x - 12 = 18$	Distributive property (a property of real numbers)
$3x - 12 + 12 = 18 + 12$	Addition property of equality
$3x = 30$	Additive inverse + additive identity
$\frac{3x}{3} = \frac{30}{3}$	Division property of equality
$x = 10$	Multiplicative inverse + multiplicative identity

Which lines above changed only *one* side? **lines 2, 4, 6**

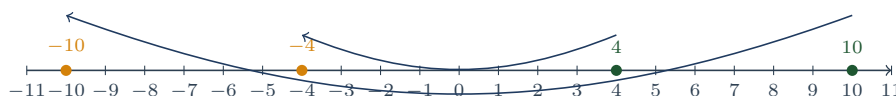
Which changed *both*? **lines 3, 5**

4. Properties of Inequality — and the One That Reverses

Everything from Section 3 carries over to $<$, $\leq$, $>$, $\geq$ — with **one** exception.

Property	Effect on the symbol
Add or subtract the same number on both sides	symbol stays the same
Multiply or divide both sides by a positive number	symbol stays the same
Multiply or divide both sides by a negative number	symbol reverses

Why does the negative one reverse? Start with something obviously true and watch what multiplying by -1 does to it on the number line.



- Start: $4 < 10$ — true, and on the line 4 sits to the **left** of 10.
- Multiply both sides by -1 : $-4 > -10$? Check the picture: -4 now sits to the **right** of -10 .
- Multiplying by a negative **reflects the entire number line through zero**. Reflection reverses **order** — so the symbol has to turn around with it.

Back to Ana and Ben. Now settle the Hook by name, not by testing.

Step	Property that allowed it
$-3x > 12$	Given
$\frac{-3x}{-3} < \frac{12}{-3}$	Division property of inequality — divided by a negative, so the symbol reverses
$x < -4$	Multiplicative inverse + multiplicative identity

So **Ben** was right. *Careful:* the symbol reverses only when you multiply or divide **by** a negative. It does **not** reverse just because the answer happens to be negative, and it does **not** reverse when you subtract.

Guided Practice

Justify each line, then answer the question underneath.

Step	Property
$-2(x + 5) \leq 6$	Given
$-2x - 10 \leq 6$	Distributive property
$-2x \leq 16$	Addition property of inequality (added 10 to both sides)
$x \geq -8$	Division property of inequality — divided by -2, so the symbol reversed

1. The symbol reversed on the last line but not the line before it. Explain the difference between what happened on those two lines. **Line 3 added 10 to both sides, and adding never changes the symbol. Line 4 divided both sides by -2 , a negative number, which reflects the number line through zero and reverses the order — so $\leq$ became $\geq$.**
2. Classify $\sqrt{16}$ into every set it belongs to. **$\sqrt{16} = 4$: natural, whole, integer, rational, real**
3. A classmate says “ $-\sqrt{4}$ must be irrational, since it has a radical *and* a negative.” Give the value and say what is actually true. **$-\sqrt{4} = -2$, which is an integer, a rational number, and a real number — not irrational. Neither the radical sign nor the negative sign decides; only the *value* does.** ..

Teacher Note

Pacing. Sections 1–2 are fast (students know the content, only the names are new). Budget the time in Sections 3–4, where the real work is the two-column justification tables — students who have only ever written bare solution steps find naming a line genuinely hard the first time. **The most valuable single line in the notes** is the boxed distinction opening Section 3: *properties of real numbers rewrite one side; properties of equality act on both*. Every subsequent lesson in Unit 1 refers back to it.

Justification wording. Accept any correct name; do not require the exact phrasing above. Some texts fold “additive inverse + additive identity” into “simplify” — accept that too, provided the student can say what was simplified and why it was legal. What is *not* acceptable is “I moved it over.”